ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** any ratio	Variable.
2	**(d)** filtration	Filter paper.
3	**(a)** mass of solute + mass of solvent	Sum.
4	**(c)** mixture of gases	Homogeneous mixture.
5	**(d)** evaporation	Water evaporates.
6	**(d)** 100 g	Sum.
7	**(a)** homogeneous mixture	Uniform.
8	**(a)** sublimation	Camphor sublimes.
9	**(d)** 10 per cent	10/100.
10	**(a)** suspension	Settles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substance made of two or more components not chemically combined (2).	2
12	Filtration (1); Distillation (1).	1+1
13	Solid changes directly to vapour (1); Camphor (1).	1+1
14	Homogeneous mixture of two or more substances (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Clear vs turbid (1); Stable vs settles (1); Cannot be filtered vs can be filtered (1).	1+1+1
16	Solution = 200 g (1); 20/200 x 100 (1); 10 per cent (1).	1+1+1
17	Composition (1); Properties (1); Separation (1).	1+1+1
	OR: Milk (1); Fog (1); Butter (1).	1+1+1
18	5/100 x 200 (1); = 10 g (1); Water = 190 g (1).	1+1+1
	OR: Hard (1); Resistant to rust (1); Better strength (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Homogeneous (1); Heterogeneous (1); Solution examples (1); Suspension examples (1); Colloids (1).	1+1+1+1+1
	OR: Homogeneous (1); Small particles (1); Stable (1); Transparent (1); No Tyndall effect (1).	1+1+1+1+1
20	Magnet (1); Water (1); Filter (1); Evaporate (1); Draw conclusion (1).	1+1+1+1+1
	OR: Spot (1); Solvent (1); Capillary action (1); Separation (1); Uses (1).	1+1+1+1+1

Mistakes to avoid:

- Calculating mass per cent using the mass of the *solvent* instead of the mass of the *solution* (solute + solvent).
- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.
- Confusing evaporation (solvent lost, solid left) with distillation (solvent collected by condensing); and using fractional distillation when the boiling points differ by more than 25 K.
- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** magnet	Magnetic.
2	**(c)** increases	Warm.
3	**(d)** colloid	Scatters light.
4	**(b)** centrifugation	Spin.
5	**(a)** no more solute	Limit.
6	**(d)** alloy	Copper and zinc.
7	**(a)** distillation	Condense vapour.
8	**(c)** more solute	Capacity.
9	**(b)** dissolves	Dissolved.
10	**(d)** chromatography	Different solubility.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Separating insoluble solid from a liquid using filter paper (2).	2
12	Solution that cannot dissolve more solute (2).	2
13	Heterogeneous mixture whose particles settle (2).	2
14	Conversion of liquid into vapour to leave dissolved solid (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Salt in water (1); Air (1); Brass (1).	1+1+1
16	Distillation (1); Fractional distillation (1); Separation of liquid air (1).	1+1+1
17	Add more solute (1); Evaporate solvent (1); Cool (1).	1+1+1
	OR: Muddy water (1); Chalk in water (1); Flour in water (1).	1+1+1
18	Cream from milk (1); Blood cells from plasma (1); Washing machines dry clothes (1).	1+1+1
	OR: Hold insoluble solid (1); Allow liquid to pass (1); Separate residue and filtrate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat the mixture (1); Vapour rises (1); Condenser cools (1); Liquid collects (1); Salt remains (1).	1+1+1+1+1
	OR: Ammonium chloride sublimes (1); Heat the mixture (1); Collect on a cool funnel (1); Salt remains (1); Separated (1).	1+1+1+1+1
20	Mixture of liquids (1); Fractionating column (1); Different boiling points (1); Crude oil (1); Air (1).	1+1+1+1+1
	OR: Sedimentation (1); Filtration (1); Chlorination (1); Safe water (1); Boiling at home (1).	1+1+1+1+1

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- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** less solute	Weak.
2	**(d)** dissolves the solute	Dissolver.
3	**(b)** separating funnel	Layers.
4	**(c)** more solute	Strong.
5	**(d)** salt solution	Aqueous solution.
6	**(a)** winnowing	Wind.
7	**(d)** pressure and cooling	Compression.
8	**(c)** colloids	Scatter.
9	**(c)** fractional distillation	Refinery.
10	**(b)** colour writing	Greek.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Scattering of light by particles in a colloid (2).	2
12	Mixture with particles that do not settle but scatter light (2).	2
13	Boiling and condensing a liquid (2).	2
14	Brass (1); Bronze (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add water and stir (1); Filter to remove sand (1); Evaporate to get salt (1).	1+1+1
16	Separate dyes (1); Test drugs (1); Forensic work (1).	1+1+1
17	Cools the vapour (1); Converts to liquid (1); Collects the distillate (1).	1+1+1
	OR: Solution = 100 g (1); 15/100 x 100 (1); 15 per cent (1).	1+1+1
18	Dissolves many substances (1); Present widely (1); Used in life (1).	1+1+1
	OR: Filter paper strip (1); Solvent (1); Sample spot (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Saturated solution (1); Cool slowly (1); Crystals (1); Filter (1); Dry (1).	1+1+1+1+1
	OR: m/m (1); v/v (1); m/v (1); Example (1); Units (1).	1+1+1+1+1
20	Particle size (1); Tyndall effect (1); No settling (1); Cannot be filtered (1); Examples (1).	1+1+1+1+1
	OR: Mixture of metals (1); Brass (1); Steel (1); Bronze (1); Uses (1).	1+1+1+1+1

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